Aman Kumar Singh

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Professional Summary

Computer Science undergraduate specializing in Full Stack Development, Artificial Intelligence, Machine Learning, and Cloud Computing. Experienced in building scalable MERN applications, RESTful APIs, and AI-powered solutions through internships, research, and hackathons. Strong foundation in Data Structures, backend engineering, and machine learning.

Education

Vellore Institute of Technology **Aug 2023 – Present**
B.Tech in CSE (CGPA: 8.55) *Chennai, Tamil Nadu*

Vig English School **March 2020 – March 2022**
Class 10 – ICSE (92.60%), *Jamshedpur, Jharkhand*
Class 12 – ISC (91.50%)

Experience

Web Dev Intern, Tata Motors **May 2025 – Jun 2025**
GitHub — Certificate *Jamshedpur*

- Built **TripOnTheGo**, a full-stack hotel booking platform using **PHP, MySQL, HTML, CSS, and JavaScript**, enabling users to search hotels by location and make reservations.
- Developed a secure admin dashboard with authentication and **CRUD** functionality to manage hotel listings, bookings, and customer records.
- Integrated PHP with a **MySQL** database for booking management and tested the application locally using **XAMPP**.

Research Intern - CCPS, VIT **May 2025 – Aug 2025**
Paper — Certificate *Hybrid*

- Developed an **Enhanced DeepLab-FusionNet** architecture with multi-scale attention mechanisms.
- Evaluated model performance on the **BUSI** and **Kvasir-SEG** datasets through benchmarking and systematic ablation studies.
- Published the research paper **"CA2PNet: A Context-Aware Multi-Scale Architecture with Adaptive Attention and Progressive Dilated Convolutions for Biomedical Image Segmentation"** in *Frontiers in Artificial Intelligence*.

Projects

AWS AI for Bharat — GitHub **Jan 2026 – Mar 2026**

- Led an Agile team to develop a multilingual AI platform for discovering **200+ government schemes**, improving development velocity by **30%**.
- Built a low-latency **RAG pipeline** using **Pinecone Vector DB**, reducing query response time by **˜35%**.
- Architected a scalable AWS backend (**API Gateway, ECS, Lambda, RDS**) achieving **99.9% uptime**.

Secure Document Cleanser — GitHub **October 2025**

- Developed a secure document processing platform using **Python, Flask, OpenCV, spaCy, and Gemini**.
- Built an automated **PII redaction pipeline** using computer vision and NLP to remove sensitive data.
- Designed a multimodal AI workflow for intelligent document summaries and actionable business insights.

Wanderlust — GitHub **April 2025**

- Developed a **full-stack accommodation booking platform** using **Node.js, Express.js, MongoDB, and EJS**, enabling users to discover, list, and reserve unique stays through a responsive interface.
- Implemented **user authentication**, CRUD operations for property listings and reviews, image uploads with **Cloudinary**, and interactive location mapping using the **Mapbox API**.
- Designed **search and filtering** functionality with a mobile-responsive UI, and deployed the application on **Render** using **MongoDB Atlas** for cloud-based data persistence and secure environment configuration.

Achievements

- **Runner-up – Yuva AI Thon** — GitHub – Built **Pixro**, a full-stack webinar hosting platform using **Next.js**, supporting live and recorded streaming, an Agentic AI virtual assistant, and responsive UI. Leveraged the App Router, optimized font loading, and modern React features to deliver a scalable and high-performance user experience.
- Secured **Silver Medal** in the **100m Sprint** (12.69s) and **Bronze Medal** in the **200m Sprint** (26.70s) at the **Inter School Athletic Meet 2018–19** organized by **Tata Motors**.

Leadership & Positions of Responsibility

Vibrance – Cultural Fest, VIT Chennai **2025**
Event Management & Registration Committee *Chennai*

Technical Skills

Programming Languages: C++, Java, Python, JavaScript, R, MATLAB, Verilog
Frontend: HTML, CSS, Bootstrap, Tailwind CSS, React.js, EJS
Backend: Node.js, Express.js, Flask, RESTful APIs
Databases: MongoDB, PostgreSQL, SQL, Pinecone Vector DB
AI/ML & Computer Vision: TensorFlow, PyTorch, Scikit-learn, spaCy, OpenCV, Tesseract OCR
Cloud & DevOps: AWS (EC2, Lambda, API Gateway, RDS, S3, CloudFront), Docker, Terraform, Render, Vercel
Libraries & Tools: Git, GitHub, PyMuPDF, python-pptx, openpyxl, Pillow

Certifications

Full Stack MERN | Cyber Security Basics | TCS iON Soft Skills | C++ Programming | C Programming | Python Programming